Unit: Unit 2: Exponents Radicals
Topic: Exponents

Name: ____________
Date: ________

Quiz (Habanero)

Q1. If the temperature of a city rises by a factor of 2^3 every 5 years, what is the total rise in temperature after 15 years?
(A) 2^9
(B) 2^6
(C) 2^3
(D) 2^1
(E) 2^0

Q2. A certain type of plant grows at a rate of 3^x, where x is the number of years. If the plant grows to 3^4 in 4 years, what is its growth rate per year?
(A) $3^{4/x}$
(B) $3^{x/4}$
(C) 3^4
(D) $3^{1/4}$
(E) $3^{1/x}$

Q3. The population of a certain species of rabbit increases by a factor of 2^2 every 2 months. If there are initially 12 rabbits, how many will there be after 6 months?
(A) 192
(B) 96
(C) 48
(D) 24
(E) 12

Q4. Simplify the expression: $(2^3)^2 \cdot 2^4$
(A) 2^{14}
(B) 2^{12}
(C) 2^{10}
(D) 2^8
(E) 2^6

Q5. What is the value of x in the equation $2^x \cdot 2^3 = 2^{10}$?
(A) 7
(B) 6
(C) 5
(D) 4
(E) 3

Q6. Simplify the expression: $(3^2)^x \cdot 3^4$
(A) 3^{2x+4}
(B) 3^{2x-4}
(C) 3^{x+4}
(D) 3^{x-4}
(E) 3^{4x}

Q7. A certain type of bacteria grows at a rate of 2^{x+1}, where x is the number of hours. If there are initially 16 bacteria, how many will there be after 3 hours?
(A) 128
(B) 64
(C) 32
(D) 16
(E) 8

Q8. What is the value of x in the equation $2^x \cdot 2^x = 2^{12}$?
(A) 6
(B) 5
(C) 4
(D) 3
(E) 2

Open Ended Questions (FRQ)

Q9. The water level in a certain lake rises by a factor of 2^x every year, where x is the amount of rainfall in inches. If the water level rises by a factor of 2^4 in 2 years, and the rainfall in the first year is 2 inches, what is the rainfall in the second year?

Q10. A certain type of algae grows at a rate of 3^{x-1}, where x is the number of days. If the algae grows to cover an area of 3^4 square meters in 5 days, what is its growth rate per day?

Chef's Tips

- Tip 1: Make sure to read the questions carefully and understand what is being asked.
- Tip 2: Use the exponent rules to simplify expressions and solve equations.
- Tip 3: Check your work and make sure your answers are reasonable.